

ff360_labs

SYNTHWAVE PRODUCTION SUITE

VHS • Neon Chorus • Midnight Reverb • Neon Width
Neon Tape Stop • Cyberpunk Glitch • RetroFX • NightDrive

Version 1.0.0 Production Release | VST3 • AU • Standalone
Windows 10/11 & macOS 12+ (Apple Silicon & Intel)

1. Suite Architecture & Shared Core

The **ff360_labs Synthwave Production Suite** is a modular collection of eight professional audio plugins built on a single, high-performance C++20 DSP engine (ff360_dsp_core). Designed for modern Synthwave, Darksynth, Cyberpunk, and Vaporwave production, every plugin shares unified visual design tokens, smooth parameter interpolation, and post-Phase-10 broadcast metering precision.

Every plugin in the suite adheres to ff360_labs' signature glassmorphic aesthetic: deep black backgrounds (#0a0a0b), matte charcoal panels, and metallic gold accents (#c9a15a). The signature macro knobs—**DEGRADE** in VHS, **PROBABILITY** in Cyberpunk Glitch, **GENERATE** in RetroFX, and **EVOLVE** in NightDrive—share an identical hero ring visual treatment and smooth parametric mapping curves.

Each plugin embeds the calibrated MeteringBridge engine, providing sample-rate independent RMS integration, 4x polyphase FIR True Peak oversampling, IEC 60268-17 VU ballistics (-4.25 VU alignment), and EBU R128 gated Loudness Range (LRA) monitoring with -70 dBFS absolute and -10 LU relative thresholds.

2. Plugin Reference & Parameter Guide

VHS models the magnetic, mechanical, and electronic imperfections of 1980s VCRs, cassette decks, and studio tape reels.

- **DEGRADE Macro:** Single-knob master control driving saturation, wow, flutter, pitch drift, and high-frequency loss along calibrated curves.
- **Saturation & Hysteresis:** Dynamic asymmetric soft-clipping modeling magnetic tape flux compression.
- **Mechanical Wow & Flutter:** Dual-rate capstan wobble (0.5 Hz wow + 6.0 Hz flutter).
- **Noise & Mechanical Hiss:** Filtered analog tape hiss and mechanical chassis hum.

A lush BBD-style chorus with stereo and 4-voice quad modes for wide synth pads and shimmering guitars.

- **Stereo / Quad Mode:** Toggle between classic 2-voice dual LFO and ultra-wide 4-voice quad delay networks.
- **Modern / Vintage Character:** Vintage mode introduces subtle clock jitter, analog warmth, and bucket-brigade high-frequency roll-off.
- **4th-Order Bass Mono:** Linkwitz-Riley low-end crossover filter ensuring 100% mono sub-bass compatibility below selectable cutoff.

Rich vintage digital reverberation featuring 6 selectable space algorithms and instant sidechain ducking.

- **6 Algorithms:** Digital Hall, Dark Plate, Gated Room (80s Phil Collins drums), Synth Room, Endless (infinite decay), and Dream (lush shimmer modulation).
- **Ducking Envelope:** Built-in single-buffer transient ducking that attenuates reverb tail during dry transient hits.
- **Infinite Freeze:** Holds the reverb buffer infinitely without overload or runaway resonance.

Precision spatial imaging engine with an integrated real-time Lissajous polar goniometer.

- **Polar Goniometer & Phase Correlation Bar:** Real-time visual feedback of stereo spread and mono compatibility.
- **Haas & Micro-Delay:** 0–25ms inter-aural time difference widening.
- **Frequency-Dependent Width:** Expands high-frequency air while preserving narrow, focused low-mids.
- **Stereo Rotation:** Panning rotation matrix rotating the stereo field without collapsing channel energy.

A dedicated slowdown and spin-up effect with selectable inertia profiles and MIDI triggering.

- **Profiles:** Vinyl Stop (turntable motor drag), Tape Stop (analog reel inertia + HF loss), and Digital Stop (stepped quantization cut).
- **Reverse Recovery:** Spins back up to speed in reverse phase for creative drops and transitions.

Rhythmic stutter generator, buffer freeze, and bitcrush processor synced to host tempo.

- **Hero Probability Knob:** Sets likelihood of glitch firing on each grid subdivision (1/4 to 1/32).
- **Reverse Playback & Freeze:** Instant reverse slice looping and infinite buffer hold.
- **Granular Pitch Shift:** -12 to +12 semitones real-time slice transposition.

On-demand synthwave transition and riser generator eliminating the need for static sample packs.

- **9 Generators:** Noise Sweep, Pitch Sweep, Laser, Reverse Swell, Impact Sub-Boom, Riser, Downlifter, Digital Sweep, Tape Sweep.
- **GENERATE Hero Control:** Instant trigger with deterministic seed recall for 100% reproducible takes.

Generative ambient backdrop engine combining analog drones, granular texture clouds, and arpeggios.

- **EVOLVE Macro:** Single master knob modulating grain drift, chorus rate, LFO speed, and shimmer depth.
- **Chord / Scale Lock:** Automatically constrains pitch generation to key (Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Synthwave Pentatonic). Supports live ChordFlow integration with automatic fallback.

3. Factory Preset Index (41 Presets)

All 41 factory presets use JSON serialization via ParameterManager for instant, glitch-free DAW recall.

Plugin	Preset Name	Category	Description & Sonic Character
VHS	Clean Master Tape	Subtle	Subtle studio reel tape saturation and gentle high-end rounding.
VHS	1984 VCR Broadcast	Vintage	Authentic 80s videotape playback with moderate wow and chassis hum.
VHS	Melancholy Cassette	Lo-Fi	Slow pitch wobble, gentle lo-fi decimation, and warm harmonic bloom.
VHS	Broken Walkman	Extreme	Severe motor drag, tape crunch, and intense wow/flutter instability.
VHS	Tape Bloom & Warmth	Harmonics	Harmonic tape drive ideal for drum buses and synthesizer master tracks.
Neon Chorus	Juno Pad Spread	Pads	Iconic 1982 analog synthesizer chorus with ultra-wide stereo movement.
Neon Chorus	Tokyo 1986 Ensemble	Vintage	4-voice quad BBD ensemble with clock drift and warm HF loss.
Neon Chorus	Dimension 4D Clean	Modern	Ultra-clean modern spatial chorus with zero phase smearing.
Midnight Reverb	Cyberpunk Cathedral	Halls	Vast metallic hall with dense reflections for dark synth backdrops.
Midnight Reverb	80s Snare Gate	Gated	Classic non-linear gated room with instantaneous explosive cutoff.
Midnight Reverb	Blade Runner Dream	Cinematic	Lush modulated shimmer reverb with slow harmonic bloom.
Neon Width	Wide Synth Pad	Pads	Haas widening + M/S expansion with safe mono sub crossover.
Neon Width	Rotating Horizon	Movement	Continuous subtle stereo rotation matrix for evolving pad movement.
Neon Tape Stop	Classic 1/2 Bar Brake	Tape	Authentic 800ms reel-to-reel motor slowdown and spin-up.
Neon Tape Stop	Vinyl Turntable Drop	Vinyl	S-curve turntable power-down with realistic needle drag.
Cyberpunk Glitch	1/16 Beat Stutter	Rhythm	Tempo-synced 1/16th rhythmic stutter repeats with 60% probability.
Cyberpunk Glitch	Dystopian Reverse	Dark	1/8th note reverse slices with -12st octave drop and bitcrush.
RetroFX	Cyberpunk 4-Bar Riser	Risers	4-bar ascending multi-saw detuned riser with resonant filter sweep.
RetroFX	Neon Laser Impact	Impacts	High-energy modulated FM laser shot followed by sub-bass drop.
NightDrive	Midnight Highway Drone	Drones	Sub-heavy analog pad drone with subtle granular raindrops.
NightDrive	Tokyo 3AM Atmosphere	Cinematic	Evolving cinematic texture in Synthwave Pentatonic scale.

4. Technical Specifications & System Requirements

- **Plugin Formats:** VST3 (64-bit), Audio Units (AUv2), Standalone Host Application.
- **Supported Operating Systems:** Windows 10 & 11 (64-bit), macOS 12 Monterey or higher (Universal Binary: Native Apple Silicon M1/M2/M3/M4 & Intel x86_64).

- **Supported Sample Rates:** 44.1 kHz, 48.0 kHz, 88.2 kHz, 96.0 kHz, 176.4 kHz, 192.0 kHz (with automatic filter coefficient warping).
- **DAW Compatibility:** Ableton Live 11/12, FL Studio 21+, Logic Pro 10.7+, Reaper 7+, Cubase 12+, Studio One 6+, Bitwig Studio 5+.
- **CPU Efficiency:** Ultra-low latency DSP core consuming < 1.4% CPU per instance at 96 kHz on modern hardware.